Info Problemset VI

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Various problems around Alpha p3-4/AIO invitational p4 difficulty as median (except maybe like 2 hard ones)

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Soccer Match II

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

Bob is currently in charge of the soccer tournament at UNSW. There have been n games played so far, and he wants to assign them each 'skill' levels. Any team with a higher skill level will beat one with a lower skill level, no 2 teams can have the same skill level, and that every team in the tournament has played a game at least once.

Input Format

The first line of input contains the integers n and k. The next n lines of input contain 2 integers each, a_i and b_i, denoting that team a_i beat team b_i.

Output Format

If there is no such possible arrangement, output '-1'. Else, output k integers, the assigned skill levels of each team.

Subtasks & Constraints

For all test cases, $0 \leq n, k \leq 10^6$.

Social Distancing II

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

The hippos of New Yorkshire are getting quite restless after multiple lockdowns and many restrictions. Bob wants to stop this, so he imposes a new type of rule. The 2 furthest apart hippos will be rewarded with desserts.

Input Format

The first line of input contains the integer n, the number of hippos. The next n lines of input contain 2 integers each, x_i and y_i, the coordinates of the ith hippo.

Output Format

Output 4 integers x_i, y_i, x_j, and y_j, the furthest 2 hippos. If there are multiple such pairs, output any of them.

Subtasks & Constraints

For all subtasks $n, x_i, y_i \geq 0$

Subtask	Points	Additional Constraints
1	10	$x_i \leq 1$ for all i, $n \leq 10^6$
2	20	$x_i \leq 10^3$ for all i, and $n \leq 10^6$
3	20	$x_i \leq 10^6$ for all i, and $n \leq 10^6$
4	50	$x_i, y_i \leq 10^9$ for all i, and $n \leq 10^6$

Hippo Marshes II

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

You are planning a trip to Sydney, and want to visit the Hippo Marshes. The Hippo Marshes are a straight-ish line of marshes, and you want to plan a trip for k days over a contiguous subarray of the marshes, while getting the maximum possible enjoyment. However, you do not know the enjoyments of each marsh, but it is guaranteed that they are all non-negative, and below 10^6, and the enjoyment trip is the sum of enjoyments of all marshes visited (each takes 1 day to visit).

Given the enjoyments of a few tourists you have talked to, find the maximum enjoyment you could get in your journey.

Input Format

The first line of input contains n, k, m, the number of hippo marshes, the length of your trip, and the number of tourists you have met.

The next m lines of input each contain 3 integers a_i, b_i, and e_i, the first and last marsh that the ith tourist visited, and the enjoyment they had from travelling.

Output Format

If the statements of the tourists cannot be all true (some may lie), output '-1', else output the maximum enjoyment you can get.

Subtasks & Constraints

For all subtasks, it is guaranteed that $n, m, k \geq 0$.

Subtask	Points	Additional Constraints
1	10	$n, m, k \leq 100$
2	20	$n, m, k \leq 1000$
3	20	$k \leq 1000$, $n, m \leq 10^6$
4	50	$n, m, k \leq 10^6$

Favourite Colour

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

Bob is painting his abstract art project, which takes place on an n by m cm canvas. He paints k rectangles, each a certain colour on the canvas.

However, Bob wants his pet Alice the Ant to also be able to enjoy this painting as much as him. However, Alice the Ant has a small problem - once she sees a new colour, she gets so amazed by it that she can never go to another colour, unless Bob removes her. Additionally, Alice the Ant is not superpowered so can only travel to different squares by across edges, not corners or teleporting/flying.

Find the maximum number of squares that Alice the Ant could travel to, given Bob's rectangle painting actions.

Input Format

The first line of input contains the integers n, m, and k. The next k lines of input contain 5 integers, x_1, x_2, y_1, and y_2, and c, denoting the left, right, top, and bottom lines of the rectangle, and the colour of the rectangle. Assume every square on the canvas is originally colour 0.

Output Format

Output a single integer, the maximum number of squares Alice the Ant could travel to.

Subtasks & Constraints

For all subtasks, $n, m, k \geq 0$

Subtask	Points	Additional Constraints
1	20	$n, m, k \leq 100$
2	30	$n, m \leq 1000$, $k \leq 10^6$, there are only 2 colours on the canvas at any time
3	50	$n, m \leq 1000$, $k \leq 10^6$

Favourite Colour II

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

Alice the Ant has escaped into the city, but is still fixated on her favourite colour. Bob is trying to find her, and is using his paint cans to help him. In the each minute, Bob moves to a different intersection using a road, and puts a colour onto it (an intersection can have multiple colours on it), cyclicly (like red, blue, red, blue...).

Your task is to determine whether Bob can travel around the cities, starting at intersection 1, such that Alice the ant at intersection a can get home to intersection b, while only using intersections with her favourite colour on them.

Input Format

The first line of input contains the integers n, m, a, b, c, and d the number of intersections, roads, Alice's position, Bob's position, the number of colours Bob has, and Alice's favourite colour, respectively.

The next m lines of input each contain 2 integers x and y, showing that there is a road from intersection x to y.

Output Format

If it is not possible for Alice to reach Bob's home, output '-1'. Else, find the minimum time for Alice to reach Bob's home, after Bob has walked around the city.

Subtasks & Constraints

For all test cases, $0 \leq n, m, a, b, d \leq 10^5$ and $0 \leq d \leq 10$

Favourite Colour II (Redux)

Time Limit: 3.52 seconds **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

Alice the Ant has escaped into the city, is still fixated on her favourite colour, but has learnt how to travel at super-speed, traversing each road in 0 time. Bob is trying to find her, and is using his paint cans to help him. In the each minute, Bob moves to a different intersection using a road, and puts a colour onto it (an intersection can have multiple colours on it), cyclicly (like red, blue, red, blue...).

Your task is to determine the minimum amount of time for Bob to move around the city such that Alice can get home (Bob still only can traverse 1 road in 1 unit).

Input Format

The first line of input contains the integers n, m, a, b, c, and d the number of intersections, roads, Alice's position, Bob's position, the number of colours Bob has, and Alice's favourite colour, respectively.

The next m lines of input each contain 2 integers x and y, showing that there is a road from intersection x to y.

Output Format

If it is not possible for Alice to reach Bob's home, output '-1'. Else, find the minimum time for Alice to reach Bob's home.

Subtasks & Constraints

For all test cases, $0 \leq n, m, a, b, d \leq 5 \times 10^4$ and $0 \leq d \leq 10$

Favourite Colour III

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

Alice and Bob have both gotten home safely after the previous incidents, and are now playing a game around the house with their friend Carl, where each room in the house is coloured a specific colour, and in each minute, Alice, Bob, and Carl will move to a different room through a door. The game ends whenever the colour of their rooms are not exactly the same. Your task is to find the maximum length of such a game.

Input Format

The first line of input contains 5 integers n, m, a, b, and c, the number of rooms, doors, and the starting positions of each player. Note that doors are only 1-way connections.

The next m lines of input each contain 2 integers x_i and y_i, showing that there is a door from room x_i to room y_i.

Output Format

If a game can go on forever, output '-1'. Else, output the maximum length of such a game, modulo $10^9 + 7$.

Subtasks & Constraints

For all subtasks, $n, m, a, b, c \geq 0$, and $n, m \leq 100$

Subtask	Points	Additional Constraints
1	10	The graph is a directed tree, with every edge pointing either to the root or leaves
2	10	The graph is a directed tree
3	20	$n, m \leq 10$
4	20	$n, m \leq 50$
5	40	No additional constraints

Winter 2

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

The prosperous kingdom of Ra~~g~~yden has been attracting other kingdoms, who want to take over the kingdom. To prevent any further attacks, he wants to minimise the routes (bidirectional) outside the city connected to his kingdom, so he can have more guards on each one.

You have been tasked to delete 0 or more routes, while making each waterhole outside Rayden able to be accessed from people inside. Additionally, you want the total sum of the lengths of all routes to be minimised, so guards can protect a more localised zone on each of them.

However, *some* citizens may think that you are stingy because of doing this, so you want to find the 2nd smallest total sum of lengths of routes kept.

Input Format

The first line of input contains 2 integers, n and m, the number of waterholes and routes. The next m lines of input each contain 3 integers a, b, and l, showing there is a route from waterhole a to waterhole b, with length l.

Output Format

Output a single integer, the 2nd smallest total sum of lengths of routes kept, while being able to travel from any waterhole to any other one. It is guaranteed that this integer exists.

Subtasks & Constraints

For all subtasks, $0 \leq n, m \leq 10^6$, and $0 \leq l \leq 10^3$.

Subtask	Points	Additional Constraints
1	10	$M = N - 1$ and you can reach every waterhole from any other one
2	15	$l \leq 1$
3	15	$n, m \leq 100$
4	25	$n, m \leq 1000$
5	35	No additional constraints

Is that my name

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

I have lost my name in a big string of letters, with no spaces! I want to find my name though, so can you help me?

Input Format

The first line of input contains a string s, my name. The second line of input is q, the number of places I want to check for where my name is.

The next q lines of input contain 1 integer a_i, meaning I want to check if my name is in the string, starting at position a_i of the string.

Output Format

Print q integers, one for each query. If my name is there, output 1, else 0.

Subtasks & Constraints

Subtask	Points	Additional Constraints
1	30	$\|s\|, q \leq 1000$
2	70	$\|s\|, q \leq 10^6$

Consistent Practice

Time Limit: 2 seconds **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

You want to get better at informatics after getting a much lower score in AIO than expected. Therefore, you want to consistently practice. Consistently practicing means that the distance between adjacent days that you practice informatics is less than or equal to k.

However, it is not as simple as practicing every day, since you also have loads of schoolwork to do. Therefore, for each day, you have assigned it an usefulness score, which may be positive or negative. Find the maximum sum usefulness scores of days you study over n days, while practicing consistently.

Input Format

The first line of input contains 2 integers n and k. The second line of input contains n integers $a_1, a_2, \ldots, a_n$, the usefulness scores of each day.

Output Format

Output a single integer, the maximum sum of usefulness scores while practicing consistently.

Subtasks & Constraints

For all subtasks, $n \geq 0$ and $0 \leq k \leq n$, the sum of all positive a_i does not exceed 10^9, and the sum of all negative a_i is not lower than -10^9.

Subtask	Points	Additional Constraints
1	10	$a_i > 0, n \leq 10^7$
2	10	$n \leq 10^3$
3	30	$n \leq 10^6$
4	50	$n \leq 10^7$

Maddy's Party II

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

Maddy has baked a very delicious pie and can't wait to throw a party and share it with the residents of Terreste Valley. However, she can only send out so many invitations, as there is currently a massive paper shortage in Terreste Valley. Therefore, she has resorted to using new online methods of social media to promote the party.

The valley has n residents that she would like to invite. Thankfully, she doesn't have to send individual invitations, as every follower of a resident will repost a flyer about the party, notifying them about it, and their followers.

The following system on social media can be described as either person i is follows all people numbered from a to b (this means person i is not very popular), or all people numbered from a to b follow person i (person i is very popular).

However, like most things today, even asking friends to post something on social media costs money! Your task is to find out that if Maddy pays person i to promote the party on social media, would person j get notified?

Input Format

The first line of input contains 3 integers, n, m, and q, the number of residents, the number of following-thingies, and the number of questions Maddy asks you.

The next m lines of input each contain 4 integers. If the first integer of the line is -1, person i is not popular, else person i is popular. The next 3 integers on each line are a, b, and i.

The next q lines of input each contain 2 integers i and j, for each of Maddy's questions.

Output Format

Output exactly q lines of output, one for each question. If the statement is true, output 'Yes', else 'No'.

Subtasks & Constraints

For all test cases, $n, m, q \geq 0$ and $m \leq n$

Subtask	Points	Additional Constraints
1	15	$n, q \leq 1000$
2	15	$a = b$ for all following-thingies, and $n, q \leq 10^5$
3	30	$i = a$ for all following-thingies, and $n, q \leq 10^5$
4	40	$n, q \leq 10^5$